MODEL QUESTION PAPER

REVISION-2010 Reg. No…………………

SUB CODE:3057 Signature………………..

THIRD SEMESTER DIPLOMA EXAMINATION IN ENGINEERING/TECHNOLOGY

DIGITAL INSTRUMENTS PRINCIPLES

(Common for EI, IT)

Maximum Marks :100 Time : 3 Hours

PART –A

(Answer All questions in one or two sentences . Each question carries *two* marks)

Marks

I

1. What is self complementary property of Excess 3 codes.
2. What is base of a number system. Give one example.
3. Distinguish between Positive and Negative Logic.
4. What is sequential Logic Circuit
5. Mention the disadvantages of Dynamic RAM

5 X 2 =10

PART -B

(Answer any five questions . Each question carries *six* marks)

II

1.Convert the following numbers to decimal number system

(a) $(11011)_2$ (b) $(370)_8$ (c) $(AF1)_{16}$

2.Explain alphanumeric codes.

3.Design a full adder using EX OR,OR,and AND Gates.

4.Define (a) Noise Margin (b) Propagation Delay Time (c) Power Dissipation

5.Explain Johnson Counter with Diagram

6.Construct SR Flipflop using NAND Gates.

7.Explain R-2R Ladder type DAC 5 X 6 =30

PART – C

(Answer one full question from each module. Each question carries *Fifteen* marks)

MODULE I

III

(a) Convert the following numbers to binary

(i) $(873)_{10}$ (ii) $(5742)_8$ (iii)$(4BAC)_{16}$ (iv) $(4.25)_{10}$ 8

(b) Reduce the expression using K Map and implement it in NAND Logic

$\sum m(0,1,4,5,6,7,9,11,15) + d\,(10,14)$ 7

OR

IV

(a) Reduce the expression $f = (B+\overline{B}C)\,(B+BC)\,(B+D)$ 4

(b) Subtract using 1's Complement and 2's Complement method

11011 – 10001 4

(c) Design an Exclusive OR function using NOR Gates only 7

MODULE II

V

(a) Draw and explain the circuit of TTL Inverter 8

(b) Design a 4 to 1 Multiplexer with it's truthtable 7

OR

VI

(a) Design a logic circuit with three inputs and one output with following condition.

"The output is high when any two ,out of three inputs are high". 8

(b) Explain Digital Comparators 7

MODULE III

VII

(a)With diagram Explain 3 bit asynchronous counter. 8

(b) What is Race around condition? How it is avoided in JK Master slave Flipflops 7

OR

VIII

(a) Explain JK Flipflop with truthtable and timing diagram 8

(b) With diagram ,explain Modulo 4 Ring Counter 7

MODULE IV

IX

(a) Write brief notes on the following

(i) ROM (ii) RAM (iii) PROM (iv) EPROM 8

(b) Explain the working of Weighted Resistor DAC 7

OR

X

(a) Explain the working of Counter RAMP type ADC 8

(b) Draw the block diagram of NVRAM and explain it’s operation 7